

# Modulus and Argument of a Complex Number



Mathematics | JEE Main & Advanced

**Don't Memorize. Visualize.**

A complex number is a point, and also an arrow from the origin. Its **modulus** is how long the arrow is. Its **argument** is which way the arrow points.

## MODULUS

How long the arrow is. Also called the magnitude.

Written  $|z|$ .

## ARGUMENT

Which way the arrow points. Also called the angle.

Written  $\arg z$ .

## A complex number is a point and an arrow

The complex number  $z = x + iy$  is the point  $(x, y)$  on the Argand plane: the real part  $x$  along the real axis (Re) and the imaginary part  $y$  along the imaginary axis (Im). Join the origin to the point and you get an arrow. For  $z = 3 + 4i$  the arrow and the two sides 3 and 4 form a right triangle.

### Modulus: how long the arrow is

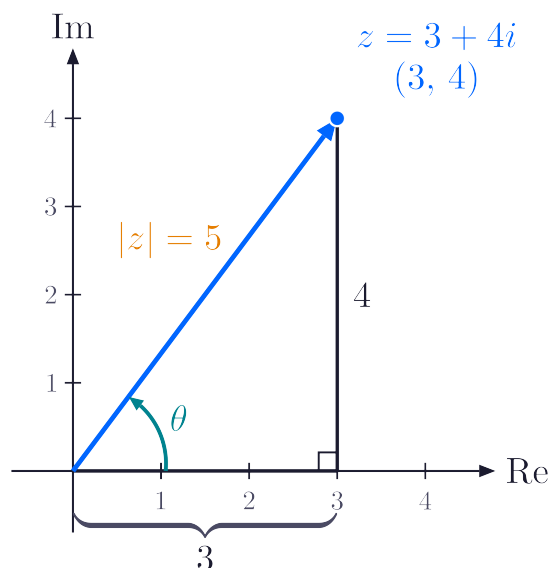
By Pythagoras,

$$|z| = \sqrt{x^2 + y^2}, \quad |3 + 4i| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$$

It is a length, so it is never negative, and it is 0 only for  $z = 0$ .

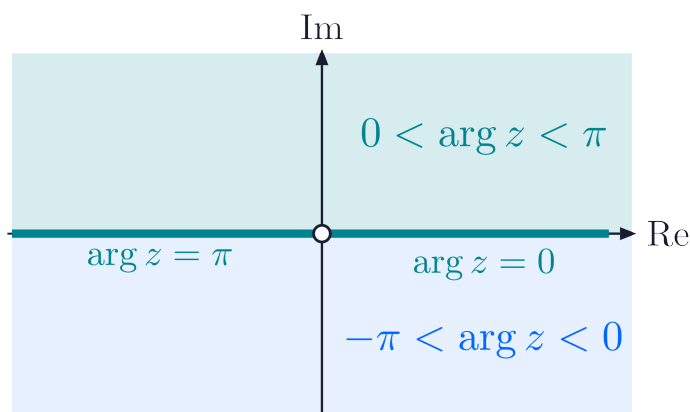
### Argument: which way the arrow points

The angle  $\arg z$  from the positive real axis to the arrow. Anticlockwise counts as positive, clockwise as negative. For  $3 + 4i$ ,  $\tan \theta = \frac{4}{3}$ , so  $\theta = \tan^{-1}\left(\frac{4}{3}\right) \approx 0.9273$  radians, about  $53.13^\circ$ . For  $z = 0$  there is no arrow, so  $\arg 0$  is not defined.

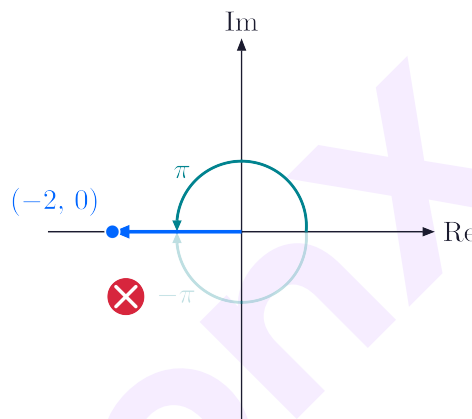


## The principal argument

One arrow has many angles: turn the other way round, or add full turns of  $2\pi$ . We agree to report one of them, the **principal argument**, in  $(-\pi, \pi]$ : the shorter turn wins.



Above the real axis the argument is between 0 and  $\pi$ , below it between  $-\pi$  and 0. The origin has no argument.



On the negative real axis both turns are half a turn. The agreement keeps the anticlockwise one:  $\arg(-2) = \pi$ , never  $-\pi$ .

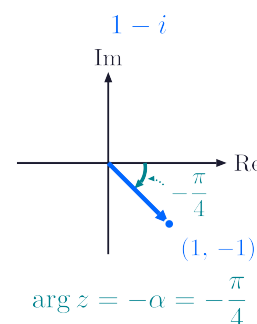
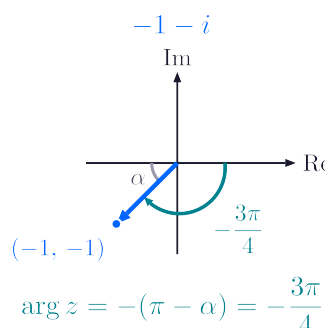
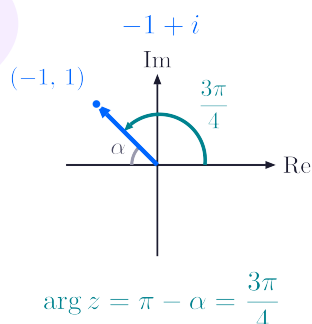
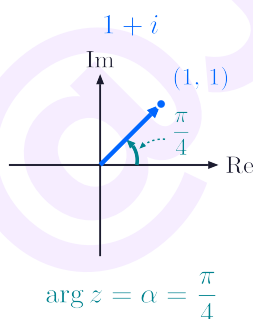
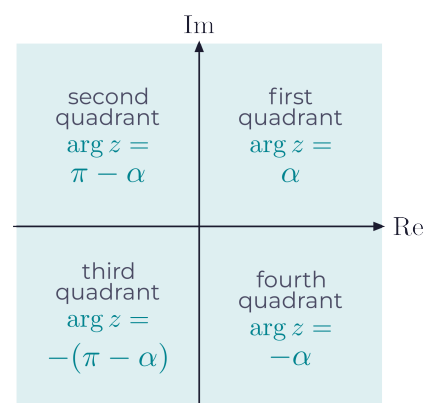
## Finding the argument in any quadrant

**STEP 1** Find the acute angle the arrow makes with the real axis,

$$\alpha = \tan^{-1} \left| \frac{y}{x} \right|.$$

**STEP 2** Read the argument from the quadrant the point is in, with the chart.

A point on an axis is in no quadrant: read its argument from the picture. It is 0 on the positive real axis,  $\pi$  on the negative real axis,  $\frac{\pi}{2}$  on the imaginary axis above the origin and  $-\frac{\pi}{2}$  below it.



The same acute angle  $\alpha = \frac{\pi}{4}$  in all four quadrants gives four different arguments.

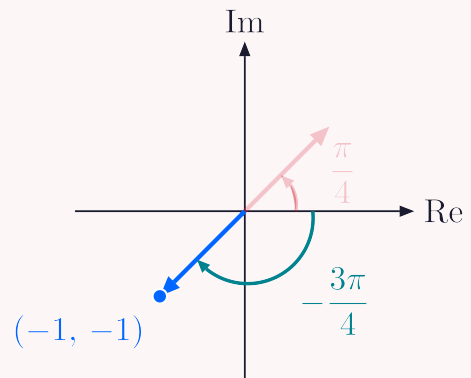
### CAREFUL

For  $z = -1 - i$  a calculator gives

$$\tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-1}{-1}\right) = \tan^{-1}(1) = \frac{\pi}{4}.$$

But an arrow at  $\frac{\pi}{4}$  points to  $(1, 1)$  in the first quadrant, and  $-1 - i$  is the point  $(-1, -1)$  in the third. Dividing  $y$  by  $x$  threw away both signs.

**The true argument is**  $-(\pi - \frac{\pi}{4}) = -\frac{3\pi}{4}$ . Always check the quadrant.



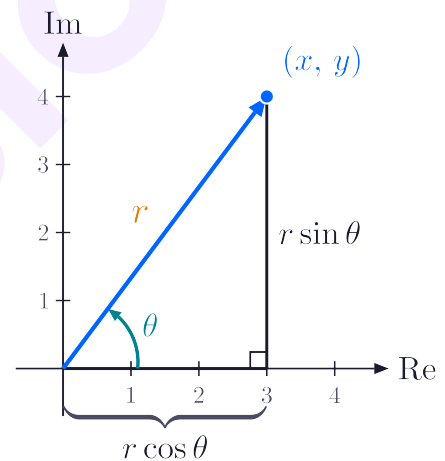
## Polar form

The two legs of the triangle are  $x = r \cos \theta$  and  $y = r \sin \theta$ , where  $r = |z|$  and  $\theta = \arg z$ . Put them into  $z = x + iy$ :

### Polar form

$$z = r(\cos \theta + i \sin \theta)$$

For our example,  $3 + 4i = 5(\cos \theta + i \sin \theta)$  with  $\theta = \tan^{-1}(\frac{4}{3})$ .



### Solved example

Find  $|z|$  and  $\arg z$  for  $z = -1 - \sqrt{3}i$ .

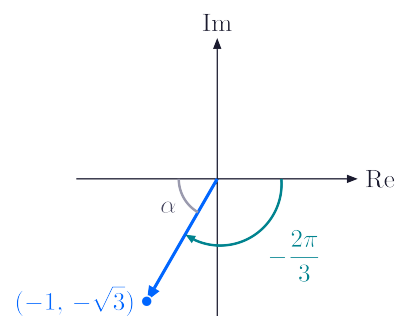
**Plot it.** The point is  $(-1, -\sqrt{3})$ , in the third quadrant.

**Modulus.**  $|z| = \sqrt{(-1)^2 + (-\sqrt{3})^2} = \sqrt{1+3} = 2$ .

**Acute angle.**  $\alpha = \tan^{-1} \left| \frac{-\sqrt{3}}{-1} \right| = \tan^{-1} \sqrt{3} = \frac{\pi}{3}$ .

**Argument.** Third quadrant, so  $\arg z = -(\pi - \frac{\pi}{3}) = -\frac{2\pi}{3}$ .

$$|z| = 2, \quad \arg z = -\frac{2\pi}{3}$$



### Key formulas

$$|z| = \sqrt{x^2 + y^2}$$

$$\alpha = \tan^{-1} \left| \frac{y}{x} \right|$$

first quadrant:  $\arg z = \alpha$

second quadrant:  $\arg z = \pi - \alpha$

third quadrant:  $\arg z = -(\pi - \alpha)$

fourth quadrant:  $\arg z = -\alpha$

principal argument:  $-\pi < \arg z \leq \pi$

polar form:  $z = r(\cos \theta + i \sin \theta)$

### Practice set

Easy to hard. Give every argument as the principal argument, in radians. Sketch each point first: the picture tells you the quadrant. The answers start on the next page.

- Q1.** Find  $|z|$  and  $\arg z$  for  $z = 1 + \sqrt{3}i$ .
- Q2.** Find  $|z|$  and  $\arg z$  for  $z = -3$ .
- Q3.** Find  $|z|$  and  $\arg z$  for  $z = -2i$ .
- Q4.** Find  $|z|$  and  $\arg z$  for  $z = \sqrt{3} - i$ .
- Q5.** Find  $|z|$  and  $\arg z$  for  $z = \frac{1+i}{1-i}$ .
- Q6.** Write  $z = -1 + \sqrt{3}i$  in polar form.
- Q7.** Find  $|z|$  and  $\arg z$  for  $z = -4 - 4i$ .
- Q8.** Let  $z = -1 - i$ . Find  $\arg(z^2)$ . Is it equal to  $2 \arg z$ ?



#### Watch the animated lesson on YouTube

Modulus and Argument of a Complex Number

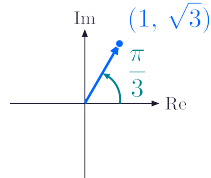
Scan the code to open the JEEVisionX channel on YouTube. This lesson is in the Complex Numbers playlist.

An Initiative by an IIT Alumnus

## Answers

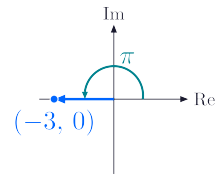
**Q1**

$|z| = 2, \arg z = \frac{\pi}{3}$   
 $|z| = \sqrt{1+3} = 2$ . First quadrant, so  $\arg z = \alpha = \tan^{-1} \sqrt{3} = \frac{\pi}{3}$ .



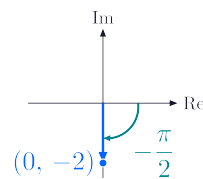
**Q2**

$|z| = 3, \arg z = \pi$   
 The point  $(-3, 0)$  is on the negative real axis. The argument is  $\pi$ , not  $-\pi$ .



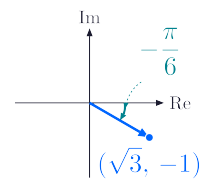
**Q3**

$|z| = 2, \arg z = -\frac{\pi}{2}$   
 The point  $(0, -2)$  is straight down the imaginary axis: a quarter turn clockwise.



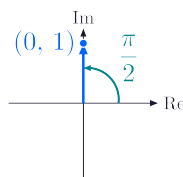
**Q4**

$|z| = 2, \arg z = -\frac{\pi}{6}$   
 $|z| = \sqrt{3+1} = 2$ . Fourth quadrant,  $\alpha = \tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}$ , so  $\arg z = -\frac{\pi}{6}$ .



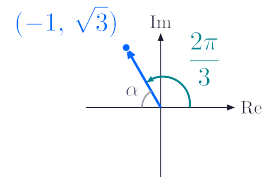
**Q5**

$|z| = 1, \arg z = \frac{\pi}{2}$   
 Multiply top and bottom by  $1+i$ :  
 $\frac{(1+i)^2}{(1-i)(1+i)} = \frac{2i}{2} = i$ , the point  $(0, 1)$ .



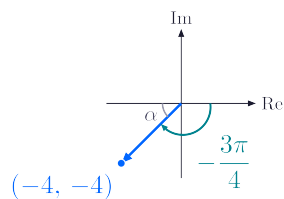
**Q6**

$z = 2 \left( \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$   
 $r = \sqrt{1+3} = 2$ . Second quadrant,  $\alpha = \frac{\pi}{3}$ , so  $\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$ .



**Q7**

$|z| = 4\sqrt{2}, \arg z = -\frac{3\pi}{4}$   
 $|z| = \sqrt{16+16} = 4\sqrt{2}$ . Third quadrant,  $\alpha = \frac{\pi}{4}$ , so  $\arg z = -(\pi - \frac{\pi}{4}) = -\frac{3\pi}{4}$ , not  $\frac{\pi}{4}$ .



**Q8**

$\arg(z^2) = \frac{\pi}{2}$ , and no.  
 $z^2 = (-1-i)^2 = 1+2i+i^2 = 2i$ , so  $\arg(z^2) = \frac{\pi}{2}$ . But  $2 \arg z = 2(-\frac{3\pi}{4}) = -\frac{3\pi}{2}$  (the grey turn), outside  $(-\pi, \pi]$ . Same direction, a full turn of  $2\pi$  apart, but only  $\frac{\pi}{2}$  is the principal argument.

